

LEPIEL, Belarus

WMO: 266590

Lat: 54.8801N

Lon: 28.6927E

Elev: 174

StdP: 99.25

Time Zone: 3.00 (E03)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.5	-18.1	-23.9	0.4	-21.2	-20.7	0.6	-17.7	7.1	-1.6	6.4	-1.1	1.0	0	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	28.7	19.8	26.8	18.9	25.1	18.1	21.0	26.7	20.0	25.2	19.0	23.5	2.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.1	23.7	18.1	13.3	22.2	17.2	12.6	21.3	61.5	26.8	57.8	25.3	54.7	23.5	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.4	4.8	DB	-25.0	31.3	2.6	1.7	-26.9	32.6	-28.4	33.5	-29.9	34.5	-31.8	35.7	(4)
(5)			WB	-25.2	22.4	2.6	1.1	-27.1	23.1	-28.6	23.8	-30.0	24.4	-31.9	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-5.0	-5.4	-0.6	7.3	12.6	16.3	18.9	17.5	12.1	6.3	1.1	-3.7	(6)	
	DBStd	9.76	5.98	6.39	4.59	4.53	4.28	3.39	3.13	3.18	3.40	3.92	4.77	5.86	(7)	
	HDD10.0	2185	464	430	328	105	22	1	0	0	18	127	266	424	(8)	
	HDD18.3	4413	723	663	587	331	185	79	31	54	188	374	516	683	(9)	
	CDD10.0	916	0	0	0	24	102	191	275	232	81	11	0	0	(10)	
	CDD18.3	102	0	0	0	0	6	19	48	28	1	0	0	0	(11)	
	CDH23.3	785	0	0	0	4	57	131	364	218	11	0	0	0	(12)	
	CDH26.7	173	0	0	0	1	8	16	93	56	1	0	0	0	(13)	
(14) Wind	WSAvg	2.2	2.5	2.4	2.5	2.2	2.2	2.1	1.8	1.8	2.0	2.3	2.5	2.6	(14)	
(15) Precipitation	PrecAvg	672	44	35	41	38	59	82	90	67	60	50	55	52	(15)	
	PrecMax	903	76	72	88	78	228	150	166	154	142	96	107	80	(16)	
	PrecMin	467	16	8	5	6	16	22	30	8	20	4	15	14	(17)	
	PrecStd	97	20	16	19	18	42	32	40	38	33	25	24	16	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	7.2	13.2	23.3	27.9	28.6	31.7	31.8	24.9	18.7	11.6	8.2	(19)	
		MCWB	5.8	4.9	7.4	13.4	18.5	19.5	21.3	21.5	16.8	13.7	10.1	7.4	(20)	
	2%	DB	4.1	4.9	10.1	20.2	24.7	26.8	29.4	27.8	21.9	15.1	9.7	6.1	(21)	
		MCWB	3.2	3.4	5.9	11.7	17.0	18.7	20.6	19.3	15.7	12.4	8.6	5.5	(22)	
	5%	DB	2.6	3.3	7.4	17.4	22.5	24.9	27.2	25.6	19.7	13.3	8.1	4.3	(23)	
		MCWB	1.8	2.3	4.6	10.5	15.0	17.8	19.4	18.3	14.5	11.3	7.1	3.7	(24)	
	10%	DB	1.5	2.0	5.4	15.0	20.2	22.9	25.2	23.6	17.8	11.8	6.9	2.9	(25)	
		MCWB	0.8	1.2	3.2	9.4	13.9	16.9	18.7	17.5	13.7	10.2	6.2	2.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	5.6	8.0	14.5	20.0	21.1	23.0	22.1	17.8	14.6	10.2	7.5	(27)	
		MCDB	6.9	7.0	12.1	21.1	25.9	25.9	29.9	29.9	23.0	17.2	11.1	8.1	(28)	
	2%	WB	3.3	3.7	6.5	12.5	17.8	19.9	21.5	20.7	16.3	12.9	8.8	5.6	(29)	
		MCDB	3.9	4.5	9.4	17.9	23.2	24.9	27.0	26.5	20.0	14.7	9.5	6.0	(30)	
	5%	WB	1.9	2.5	4.9	11.4	16.2	18.7	20.2	19.5	15.3	11.6	7.3	3.7	(31)	
		MCDB	2.5	3.2	7.0	15.9	21.1	23.3	25.7	23.6	18.4	13.0	7.9	4.2	(32)	
	10%	WB	1.0	1.3	3.6	10.2	14.7	17.6	19.3	18.3	14.3	10.4	6.2	2.2	(33)	
		MCDB	1.4	1.9	5.1	14.4	18.9	21.4	24.2	22.0	17.1	11.8	6.8	2.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	5.6	6.7	8.8	9.4	8.7	9.0	9.0	8.3	5.9	3.9	3.9	(35)	
		MCDBR	3.8	4.6	9.1	12.7	12.9	11.6	11.9	12.4	11.8	8.3	5.0	3.5	(36)	
	5% WB	MCWBR	3.6	3.8	5.9	6.1	5.8	5.1	4.7	4.9	6.2	5.5	4.3	3.1	(37)	
		MCWBR	3.4	4.4	8.0	10.6	11.2	9.9	10.7	10.4	9.6	7.2	4.4	3.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.287	0.321	0.351	0.402	0.393	0.379	0.407	0.411	0.372	0.349	0.329	0.289	(40)	
		taud	2.158	2.117	2.164	2.209	2.315	2.377	2.312	2.296	2.374	2.403	2.265	2.185	(41)	
	Ebn at Noon		629	735	808	817	851	867	832	802	788	716	577	539	(42)	
		Edh at Noon	62	93	112	125	119	113	118	113	92	71	58	49	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.60	1.47	2.80	3.98	5.14	5.58	5.27	4.37	2.91	1.40	0.48	0.35	(44)	
	RadStd		0.14	0.28	0.45	0.49	0.54	0.48	0.53	0.42	0.22	0.28	0.09	0.10	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (9 neighbors)		+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+93	N/A	

Nomenclature: See separate page